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Top Level

File: top_level.kicad_sch

References

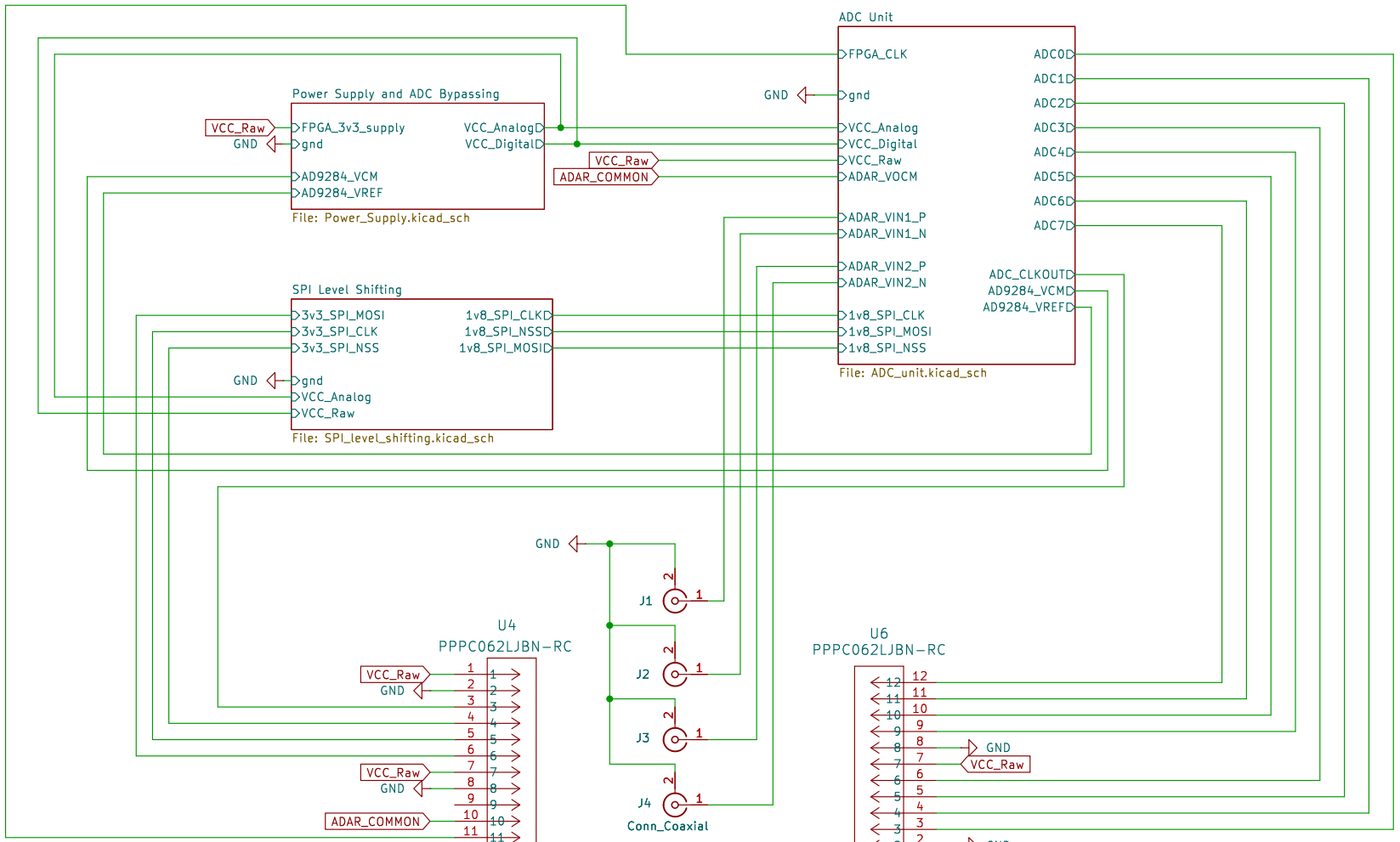
File: References.kicad_sch

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Sheet: /
File: AD9248.kicad_sch

Title: Table of Contents

Size: A4	Date: 2026-10-25	**Rev: 1**
KiCad E.D.A. 9.0.8		Id: 1/6



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Sheet: /Top Level/
File: top_level.kicad_sch

Title: Top Level ADC

Size: A4	Date: 2026-10-25	**Rev: 1**
KiCad E.D.A. 9.0.8		Id: 2/6

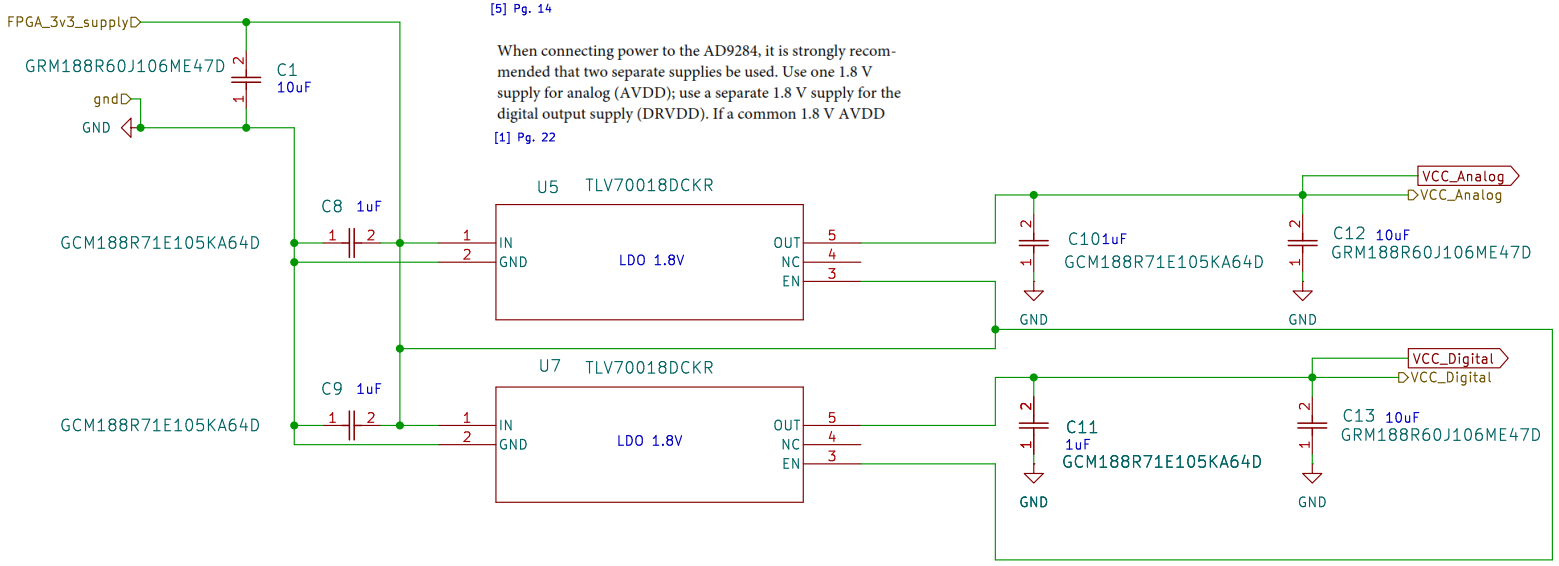
AD9248 POWER SUPPLY AND DECOUPLING

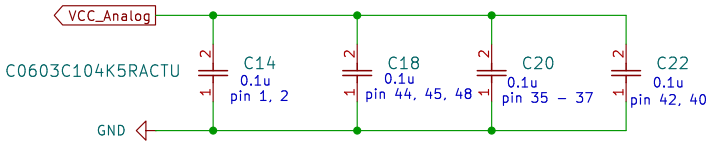
Although an input capacitor is not required for stability, it is good analog design practice to connect a 0.1-µF to 1-µF, low ESR capacitor across the IN pin and GND in of the regulator. This capacitor counteracts reactive input

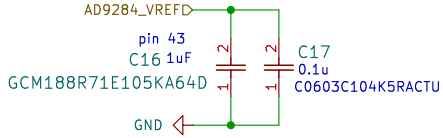
[5] Pg. 14

When connecting power to the AD9284, it is strongly recommended that two separate supplies be used. Use one 1.8 V supply for analog (AVDD); use a separate 1.8 V supply for the digital output supply (DRVDD). If a common 1.8 V AVDD

[1] Pg. 22



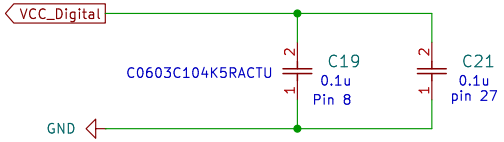


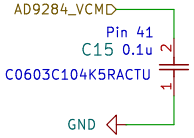


Reference Decoupling

Decouple the VREF pin externally to ground with a low ESR, 1.0 µF capacitor in parallel with a low ESR, 0.1 µF ceramic capacitor.

[1] Pg. 22





VCM

The VCM pin should be decoupled to ground with a 0.1 µF capacitor.

[1] Pg. 22

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Sheet: /Top Level/Power Supply and ADC Bypassing/
File: Power_Supply.kicad_sch

Title: Power Supply and AD9248 Bypassing

Size: A4	Date: 2026-10-25	**Rev: 1**
KiCad E.D.A. 9.0.8		Id: 3/6

C6 0.001u
C0603C102K5RACTU
C7 0.1u
C0603C104K5RACTU
U2
DS90LV048ATMX/NOPB
GND
VCC_Raw
-EN
EN
ADC0
ADC1
ADC2
ADC3
ROUT4
ROUT3
ROUT2
ROUT1
RIN4−
RIN4+
RIN3−
RIN3+
RIN2−
RIN2+
RIN1−
RIN1+

C4 0.001u
C0603C102K5RACTU
C5 0.1u
C0603C104K5RACTU
U3
DS90LV048ATMX/NOPB
ADC_CLKOUT

C2 0.001u
C0603C102K5RACTU
C3 0.1u
C0603C104K5RACTU
U1
DS90LV048ATMX/NOPB
ADC4
ADC5
ADC6
ADC7

U15 AD8052ARMZ−REEL7
C0603C104K5RACTU C31 1n
330 ohm U14
ADAR_VIN1_P
U16 330 ohm
C35 1n
C0603C104K5RACTU
ADAR_VOCM
OUT1
−IN1
+IN1
−VS
+VS
OUT2
−IN2
+IN2
1n C30
C0603C104K5RACTU
330 ohm U17
U9 330 ohm
ADAR_VIN1_N
1n C36
C0603C104K5RACTU
ADAR_VOCM

U20 AD8052ARMZ−REEL7
C0603C104K5RACTU C37 1n
330 ohm U18
ADAR_VIN2_P
U19 330 ohm
C38 1n
C0603C104K5RACTU
ADAR_VOCM
1n C40
C0603C104K5RACTU
330 ohm U21
U22 330 ohm
ADAR_VIN2_N
1n C39
C0603C104K5RACTU
ADAR_VOCM

33 ohm R12
RC1206FR−0733RL
R13
R14
33 ohm
R15
4.7p C41
06031AR47CAT2A
C42
4.7p

VCC_Analog
VCC_Digital
VCC_Raw
gnd
GND

RBIAS

The AD9284 requires that a 10 kΩ resistor be placed between the RBIAS pin and ground. This resistor, which sets the master current reference of the ADC core, should have at least a 1% tolerance.

[1] Pg. 22

R8 10k
CRCW060310K0FKEA

ADC U8
AD9284BCPZ−250
AVDD
DNC
RBIAS
DRGND
DRVDD
D0−(LSB)
D0+(LSB)
D1−
D1+
D2−
D2+
D3−
D3+
DCO−
DCO+
D4−
D4+
D5−
D5+
D6−
D6+
D7+(MSB)
D7−(MSB)
EPAD
HTAB
VIN−B
VIN+B
VIN+A
VIN−A
VREF
VCM
CLK+
CLK−
CSB
SDIO/PWDN
SCLK
*OE
AD9284_VREF
AD9284_VCM
1v8_SPI_NSS
1v8_SPI_MOSI
1v8_SPI_CLK
OE
AVDD (default)
AGND (default)
Outputs in high impedance
Outputs enabled

[1] Pg. 17

R1 R2 R3 R4 R5 R7 R9 R10
RMCF1206FG100R
100 ohm

Mandatory 100 ohm differential impedance prior to reciever input

C0603C104K5RACTU C28 0.1u
C29 0.1u
R11 100 ohm
RMCF1206FG100R

U13
DS90LV011ATMF/NOPB
FPGA_CLK
TTLIN
VDD
OUT+
OUT−
C23 0.1u
C0603C104K5RACTU
C24 0.001u
C0603C102K5RACTU

AD951x LVDS DRIVER
100Ω
0.1µF
CLK+
CLK−
ADC

[1] Pg. 14

9.2 Typical Application

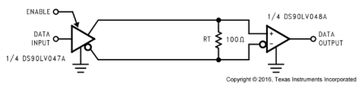


Figure 19. Balanced System Point-to-Point Application

9.2.1 Design Requirements

When using LVDS devices, it is important to remember to specify controlled impedance PCB traces, cable assemblies, and connectors. All components of the transmission media must have a matched differential impedance of about 100 Ω. They must not introduce major impedance discontinuities.

[4] Pg. 12

100 ohm R6

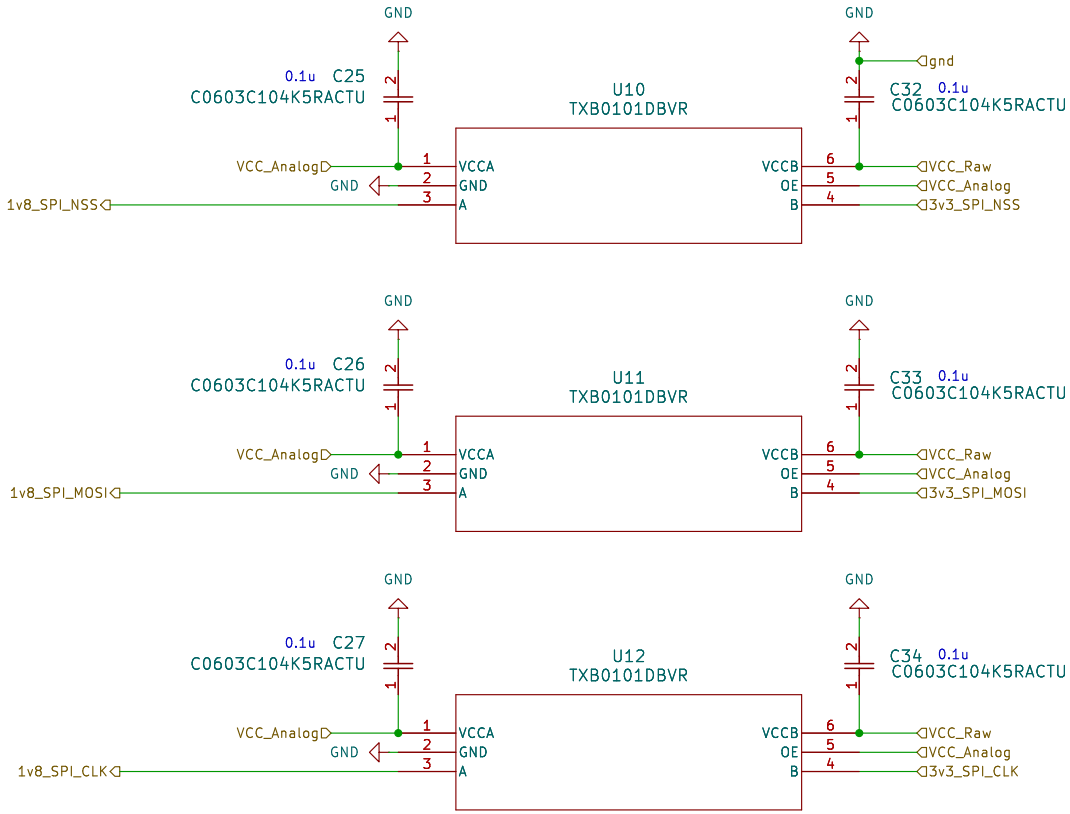
Bypass capacitors must be used on power pins. Use high-frequency ceramic (surface mount is recommended) 0.1-µF and 0.001-µF capacitors in parallel at the power supply pin with the smallest value capacitor closest to the

[4] Pg. 13

Supply voltage (V_{CC})	–0.3	4	V
Input voltage (R_{IN+}, R_{IN-})	–0.3	3.6	V
Enable input voltage (EN, EN*)	–0.3	V_{CC} + 0.3	V
Output voltage (R_{OUT})	–0.3	V_{CC} + 0.3	V

[4] Pg. 3

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Sheet: /Top Level/ADC Unit/
File: ADC_unit.kicad_sch
Title: ADC Module
Size: A3 | Date: 2026−10−25 | **Rev: 1**
KiCad E.D.A. 9.0.8 | Id: 4/6



SDIO/PWDN	AVDD (default)	Chip in full power-down
	AGND	Normal operation

[1] Pg. 17

SPI Port

The SPI port should not be active during periods when the full dynamic performance of the converter is required. Because the SCLK, CSB, and SDIO signals are typically asynchronous to the ADC clock, noise from these signals can degrade converter performance. If the on-board SPI bus is used for other devices, it may be necessary to provide buffers between this bus and the AD9284 to prevent these signals from transitioning at the converter inputs during critical sampling periods.

[1] Pg. 22

Note: MOSI needs to be pulled low
post configuration, and
CSB/NSS should be held high
CLK should be inactive

6.3.3 Enable and Disable

The TXB0101 has an OE input that is used to disable the device by setting OE = low, which places all I/Os in the high-impedance (Hi-Z) state. The disable time (t_{dis}) indicates the delay between when OE goes low and when the outputs are actually disabled (Hi-Z). The enable time (t_{en}) indicates the amount of time the user must allow for the one-shot circuitry to become operational after OE is taken high.

[6] Pg. 17

6.3.2 Power Up

During operation, make sure that $V_{CCA} \leq V_{CCB}$ at all times.

[6] Pg. 17

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Sheet: /Top Level/SPI Level Shifting/
File: SPI_level_shifting.kicad_sch

Title: SPI Level Shifting

Size: A4	Date: 2026-10-25	**Rev: 1**
KiCad E.D.A. 9.0.8		Id: 6/6

DATASHEETS, USER MANUALS, AND REFERENCES

[1] AD9284 8 bit ADC datasheet: https://www.analog.com/media/en/technical-documentation/data-sheets/ad9284.pdf

[2] Basys 3 Reference Manual : https://digilent.com/reference/programmable-logic/basys-3/reference-manual

[3] FPGA (Artix 7 XC7A35T) Datasheet : https://docs.amd.com/v/u/en-US/ds181_Artix_7_Data_Sheet

[4] DS90LV048A LVDS driver : https://www.ti.com/lit/ds/symlink/ds90lv048a.pdf?ts=1792865808432

[5] TLV700 LDO : https://www.ti.com/lit/ds/symlink/tlv700.pdf?ts=1792804791454

[6] TXB0101 Level Shifter : https://www.ti.com/lit/ds/symlink/tlv700.pdf?ts=1792804791454

Sheet: /References/		
File: References.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.8		Id: 5/6